Out-of-Sample Forecasts of Quarterly EPS for Chinese A-Shares: Seasonal Random Walk, Foster, and XGBoost, 2005–2021

Forecast-accuracy report for the PEAD / gambler's-fallacy project

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Abstract

Standardized unexpected earnings (SUE) are only as good as the expectation model behind them. Using 127,454 Shanghai and Shenzhen firm-quarter announcements from FY2005Q1 to FY2021Q1, I compare four expectation models for quarterly EPS in a strictly out-of-sample design. The models are the seasonal random walk (SRW), the SRW with drift, a Foster (1977) seasonal autoregression, and a pooled XGBoost model. All share a 12-quarter rolling window and the information set available the day before each announcement. On the common sample of 95,732 forecasts, XGBoost lowers MAE by 20.8% and RMSE by 18.5% relative to the SRW. It is the most accurate model by MAE in every year and beats the SRW for 87.7% of firms. The firm-specific drift and Foster models are *less* accurate than the SRW: short Chinese EPS histories cannot estimate the extra parameters precisely enough.

1 Introduction and motivation

Post-earnings-announcement drift (PEAD) is measured by sorting firms on an earnings surprise, so the surprise has to approximate what investors expected. The workhorse measure scales the seasonal change in quarterly EPS, $UE_{i,q} = EPS_{i,q} - EPS_{i,q-4}$, by its historical standard deviation. That is, it uses a seasonal random walk as the expectation (Bernard and Thomas, 1990). Foster (1977) showed that seasonal differences of quarterly earnings are positively autocorrelated, so a seasonal autoregression can beat the pure random walk. Bernard and Thomas (1990) built their PEAD evidence on exactly this autocorrelation structure. If a better expectation model is available, the SRW leaves predictable components in "unexpected" earnings. Surprise sorts then become noisier, and PEAD estimates may be attenuated or mis-attributed.

Two recent strands make the choice of expectation model more pressing. First, machine-learning (ML) forecasts can be more accurate than traditional benchmarks, but conclusions about their superiority depend on a clean, look-ahead-free training design and on what the forecast is meant to represent (Zhang et al., 2025). Second, Campbell et al. (2026) stress that the most accurate statistical forecast is not automatically the market's expectation. Which model to use depends on whether a study needs an accurate forecast or a proxy for investors' beliefs. In the same spirit, Pagach and Warr (2020) revisit the maintained assumption that analysts dominate time-series models for quarterly earnings. For Chinese A-shares, where analyst coverage was thin for much of the sample, time-series and ML forecasts are the natural candidates.

This report is limited to forecast accuracy. It asks which of four models best predicts quarterly EPS out of sample, how stable the ranking is over time, and whether the gains are broad-based across firms. The saved forecasts give alternative surprises $UE^{(m)}_{i,q} = EPS_{i,q} - \widehat{EPS}^{(m)}_{i,q}$ for the PEAD analysis.

2 Data and methodology

Data reconstruction. The only input is the monthly Shanghai–Shenzhen panel (515,915 firm-months, 3,594 firms). Its quarterly fields are stored by calendar quarter rather than by information arrival. Each calendar year's rows carry that fiscal year's four EPS values, and `eps_q`k/`date_q`k sit in the three rows of calendar quarter k. A January row therefore already contains Q1 EPS that is published in April. I rebuild the data at the firm–fiscal-quarter level and treat a value as known only from its publication date `date_q`k. Three checks shape this step:

- *Standalone values.* EPS is standalone quarterly, not year-to-date: the median ratio of Q2 (Q4) to Q1 EPS is about 1.3 (1.2–1.6), and only 5–15% of profitable firm-years rise monotonically through the year.
- *Adjusted values.* EPS is retroactively adjusted for later bonus issues, which keeps year-over-year changes on a common share basis.
- *Unreliable dates.* Publication dates before FY2004 are mostly vendor backfill dates (Nov 2003 – Jan 2004). They are used as given, which is conservative. Values with missing or impossible dates are never used, and identical values repeated across fiscal years are dropped.

Evaluation targets are all firm-quarters of FY2005Q1–FY2021Q1 (FY2021: Q1 only, since the panel ends in May 2021) announced within 365 days of quarter end. Table 1 reports coverage.

Information order. The fiscal order of quarters is not the order in which information arrives. Annual (Q4) and Q1 reports share an April 30 deadline, and the prior-year annual report was released on the *same day* as the Q1 report for 4% of Q1 announcements in 2005–2012 but 43% in 2019–2021 (notebook Figure 0). For these targets EPS_{q-1} is not public at the forecast origin, and every model below is built to respect this.

Models. Let $\Delta_4 EPS_s = EPS_s - EPS_{s-4}$.

M1 *SRW:* $\widehat{EPS}_q = EPS_{q-4}$.

M2 *SRW with drift:* $\widehat{EPS}_q = EPS_{q-4} + \hat{\mu}_i$, where $\hat{\mu}_i$ is the mean of the firm's $\Delta_4 EPS_s$ in the estimation window (≥ 4 changes required).

M3 *Foster seasonal AR(1):* firm-specific OLS of $\Delta_4 EPS_s = \alpha_i + \phi_i \Delta_4 EPS_{s-1} + \varepsilon_s$ (≥ 8 pairs required, $\hat{\phi}_i$ capped to $[-1, 1]$). The forecast is $EPS_{q-4} + \hat{\alpha}_i + \hat{\phi}_i \Delta_4 EPS_{q-1}$. If EPS_{q-1} is not yet public, the iterated two-step form $EPS_{q-4} + \hat{\alpha}_i(1 + \hat{\phi}_i) + \hat{\phi}_i^2 \Delta_4 EPS_{q-2}$ is used. As a robustness check, ϕ is pooled across firms (within-firm estimator on firm-scaled, winsorized pairs), keeping a firm-specific intercept.

M4 *XGBoost* (Chen and Guestrin, 2016): one model pooled across firms, predicting $\Delta_4 EPS_q$ scaled by the firm's mean $|EPS|$ over its last eight known quarters. Features: scaled EPS lags 1–8 and 12, YoY-change lags 1–4, window mean/median/SD of YoY changes, EPS volatility, trailing-four-quarter growth, loss frequency and a lag-4 loss indicator, the log EPS scale, history length, the fiscal quarter, and the number of quarters to the latest public EPS. Lags not yet public are left missing. The grid is deliberately small (depth 2–4, learning rate 0.05, subsampling 0.8, early stopping).

Rolling out-of-sample design. All models use the same origin, horizon and window:

- *Origin and horizon.* The forecast is made the day before the target's announcement date A. Inputs are restricted to EPS announced strictly before A, and the horizon is the next announcement.
- *Window.* Parameters are estimated on training targets in fiscal quarters $q-12, \ldots, q-1$ (a 12-quarter rolling window) that were announced before the first day of A's month. This common estimation cutoff lets XGBoost be refit monthly (129 fits for 232 re-estimation batches). Twelve

quarters is also the longest window feasible from 2005, since Q1 and Q3 data are sparse before FY2002.

- *XGBoost training and tuning.* XGBoost training rows are built "as of" their own announcement dates, and the training target is winsorized with training-set quantiles. Hyperparameters are re-tuned each January using only data available then: training targets announced in the previous year form the validation set, and earlier ones the fit set.

Leakage controls. The notebook verifies four things:

1. every input and estimation datum predates its cutoff;
2. a slow, label-based re-implementation reproduces M1–M3;
3. an end-to-end perturbation test holds: replacing all EPS announced after 1 January 2008 with random numbers leaves all 12,303 earlier forecasts bit-identical, including XGBoost tuning and refits, while a deliberately leaky variant fails the same test;
4. the M1 surprises reproduce the existing Stata UE measure for 100.0% of 74,390 matched announcements.

Evaluation. Accuracy is measured by MAE and RMSE in CNY per (adjusted) share on the identical common sample where all four models are available. Improvements are $\%\Delta MAE_m = 100(1-MAE_m/MAE_{SRW})$, and likewise for RMSE. Paired tests use the per-forecast loss differential $d = L(e_{SRW}) - L(e_m)$. They report (i) t-statistics with standard errors clustered by firm and fiscal quarter (Cameron et al., 2011; Thompson, 2011), and (ii) Diebold–Mariano statistics on the quarterly mean differential with Newey–West errors (Diebold and Mariano, 1995; Newey and West, 1987).

Table 1: Sample construction and model availability. A target is a firm-quarter announcement in FY2005Q1–FY2021Q1. M3's minimum of eight Foster pairs is the binding requirement for the common sample.

	Firm-quarters	% of targets	Firms
Evaluation targets (FY2005Q1–FY2021Q1)	127,454	100.0	3,383
M1 SRW: EPS_{q-4} public before A	117,902	92.5	3,120
M2 SRW + drift: $\geq$4 YoY changes	109,129	85.6	2,970
M3 Foster: $\geq$8 pairs	95,732	75.1	2,751
M4 XGBoost	117,902	92.5	3,120
Foster, pooled ϕ: $\geq$4 pairs	105,786	83.0	2,923
Common sample (M1–M4 all available)	95,732	75.1	2,751
Firms with $\geq$12 common-sample forecasts	92,584	72.6	2,124

3 Overall results

Table 2 reports the main comparison. XGBoost is the only model that beats the SRW. Its MAE of 0.0681 against 0.0860 is a 20.8% improvement, and its RMSE improvement is 18.5%. The improvement is also visible in the median absolute error (0.0225 vs. 0.0250) and in scale-free errors (14.0%). The added complexity of the parametric models does not pay off. Drift raises MAE by −8.4%, and firm-specific Foster by −14.6%. Even with ϕ pooled across firms, Foster stays below the SRW (−7.5%). The estimated persistence of seasonal differences is low (median firm $\hat{\phi} = 0.08$, median pooled $\phi = 0.12$), so estimation noise in $\hat{\alpha}_i$ and $\hat{\phi}_i$ outweighs the small predictable component.

Extreme errors. Full-sample RMSE is dominated by a handful of implausible EPS values. For example, one firm (ANTONG HOLDINGS) reports quarterly EPS of 27.8, 36.5 and −64.1 CNY in 2016, which produces the six largest Foster errors. Accordingly, XGBoost's RMSE gain is only

Table 2: Forecast accuracy. MAE, RMSE and MedAE are in CNY per share. %Δ is the improvement relative to the SRW (positive = better); sMAE scales each absolute error by the firm's historical EPS scale. Panel B removes the 1% of forecasts with the largest absolute error of *any* model. Stars: significance of the mean loss differential, two-way clustered by firm and fiscal quarter (*** 1%, ** 5%, * 10%).

	MAE	RMSE	%ΔMAE	%ΔRMSE	MedAE	%ΔsMAE
Panel A: common sample (N = 95,732)						
SRW	0.0860	0.4791	–	–	0.0250	–
SRW + drift	0.0933	0.5006	−8.4***	−4.5*	0.0287	−9.5
Foster	0.0986	0.6211	−14.6***	−29.6	0.0293	−11.8
XGBoost	0.0681	0.3906	20.8***	18.5*	0.0225	14.0
Panel B: trimmed common sample (N = 94,774)						
SRW	0.0641	0.1328	–	–	0.0250	–
SRW + drift	0.0703	0.1415	−9.6***	−6.6***	0.0282	−10.4
Foster	0.0727	0.1460	−13.4***	−10.0***	0.0288	−12.8
XGBoost	0.0527	0.1093	17.8***	17.7***	0.0221	14.3
Panel C: model-specific samples (each vs. SRW on the same forecasts; last column = N)						
SRW	0.0810	0.4411	–	–	0.0240	117,902
SRW + drift	0.0906	0.4761	−8.6	−4.6	0.0280	109,129
Foster	0.0986	0.6211	−14.6	−29.6	0.0293	95,732
XGBoost	0.0648	0.3598	20.1	18.4	0.0216	117,902
Panel D: robustness, Foster with pooled ϕ (common sample)						
Foster, pooled ϕ	0.0925	0.5091	−7.5	−6.3	0.0282	−8.0

marginally significant in the full sample ($t = 1.77$; Table 3). After symmetric trimming of the 1% largest errors (Panel B), XGBoost improves MAE by 17.8% and RMSE by 17.7%, and both differentials are highly significant ($t = 9.07$ for squared loss). XGBoost has the lower average absolute loss in 95.4% of the 65 fiscal quarters. The gain is concentrated where surprises are large. Forecasts in the top decile of $|\Delta_4 EPS|$ account for 95.0% of XGBoost's net MAE reduction, while for near-zero changes, where the SRW is almost right by construction, XGBoost is slightly worse. This conditioning on the benchmark's own ex-post error favours the SRW in the low buckets, so the pattern is descriptive.

Table 3: Paired loss-differential tests, $d = L(e_{SRW}) - L(e_m)$ ($d > 0$: model beats SRW). $t_{\text{firm, quarter}}$: two-way clustered; t_{DM} and p_{DM}: Diebold–Mariano test on quarterly mean differentials (Newey–West, 4 lags).

Model	Loss	$\bar{d}$	$t_{\text{firm, quarter}}$	t_{DM}	p_{DM}	% quarters $d > 0$
Common sample						
SRW + drift	absolute	−0.0072	−12.17	−11.17	0.000	0.0
	squared	−0.0211	−1.73	−2.72	0.006	15.4
Foster	absolute	−0.0125	−6.20	−9.64	0.000	0.0
	squared	−0.1563	−1.13	−1.49	0.137	10.8
XGBoost	absolute	0.0179	7.79	8.79	0.000	95.4
	squared	0.0769	1.77	1.74	0.081	89.2
Trimmed sample						
SRW + drift	absolute	−0.0061	−13.71	−12.64	0.000	0.0
	squared	−0.0024	−10.65	−9.57	0.000	3.1
Foster	absolute	−0.0086	−9.41	−12.36	0.000	1.5
	squared	−0.0037	−9.60	−10.50	0.000	1.5
XGBoost	absolute	0.0114	11.72	10.34	0.000	96.9
	squared	0.0057	9.07	10.01	0.000	96.9

4 Time variation

Figure 1 and Table 4 show that the ranking is stable. XGBoost has the lowest MAE in 17 of 17 fiscal years, with annual improvements between 11.6% and 35.6% and no trend: 21.0% in 2005–2009, 22.7% in 2010–2014 and 20.2% in 2015–2021Q1. It has the lowest raw RMSE in 16 years. The exception, 2016, is the ANTONG episode; on the trimmed sample, XGBoost is best by RMSE in every year (Figure 2b). The drift and Foster models are worse than the SRW in every year by MAE. All models' errors rise after 2015 as the cross-section widens, but the relative ordering is unchanged. The edge is smallest for Q1 targets whose prior annual report was not yet public (7.5% vs. 13.6% when it was; by-quarter results are in `output/tables/tab6_quarter.tex`). For Q1 targets with a known Q4, Foster is particularly poor (−56.4%), because it extrapolates noisy year-end seasonal changes into Q1.

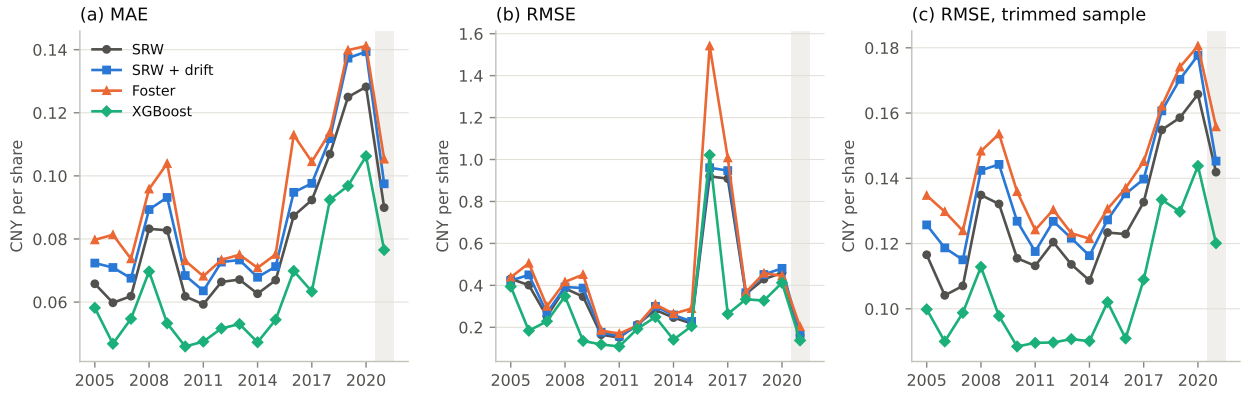


Figure 1: Annual forecast accuracy on the common sample (CNY per share), by target fiscal year. Panel (c) uses the symmetrically trimmed sample. The shaded year 2021 contains Q1 targets only.

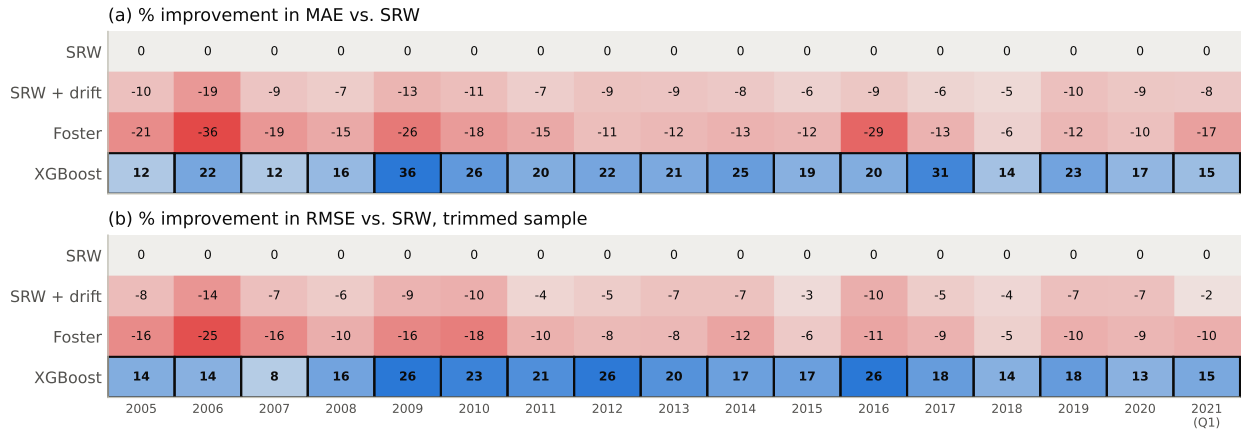


Figure 2: Percentage improvement over the SRW by fiscal year (blue = better, red = worse). The outlined, bold cell is the best model in each year. (a) MAE, common sample; (b) RMSE, trimmed sample.

Table 4: Annual accuracy. SRW columns are MAE/RMSE in CNY per share; model columns are %Δ vs. SRW. Stars on %ΔMAE use firm-clustered standard errors within the year. XGB^{trim}: RMSE improvement on the trimmed sample.

		MAE				RMSE					Best model	
FY	N	SRW	Drift	Foster	XGB	SRW	Drift	Foster	XGB	XGB^{trim}	MAE	RMSE
2005	2,282	0.0658	−10.0***	−21.2***	11.6***	0.4311	2.0	−1.6	8.5	14.3	XGBoost	XGBoost
2006	3,101	0.0598	−18.7***	−36.1***	21.7**	0.4011	−12.2	−26.0	54.2	13.6	XGBoost	XGBoost
2007	3,577	0.0619	−9.2***	−19.1***	11.6***	0.2522	−6.6	−18.8	9.3	7.7	XGBoost	XGBoost
2008	4,001	0.0832	−7.3***	−15.1***	16.3***	0.3857	−1.7	−8.1	9.8	16.3	XGBoost	XGBoost
2009	4,105	0.0827	−12.6***	−25.6***	35.6***	0.3461	−11.8	−30.4	61.0	26.0	XGBoost	XGBoost
2010	4,552	0.0618	−10.7***	−18.3***	25.7***	0.1648	−6.9	−12.3	28.2	23.4	XGBoost	XGBoost
2011	5,016	0.0593	−7.2***	−15.1***	19.9***	0.1514	−4.1	−12.0	28.2	20.9	XGBoost	XGBoost
2012	5,200	0.0664	−9.5***	−10.5***	22.2***	0.2129	2.8	1.9	9.5	25.5	XGBoost	XGBoost
2013	5,806	0.0671	−9.3***	−11.8***	21.0***	0.2825	−6.1	−9.4	11.7	20.1	XGBoost	XGBoost
2014	6,791	0.0626	−8.3***	−13.1***	24.6***	0.2464	−5.1	−7.5	42.8	17.1	XGBoost	XGBoost
2015	7,510	0.0670	−6.4***	−12.2***	18.8***	0.2182	−4.5	−32.7	5.8	17.3	XGBoost	XGBoost
2016	7,816	0.0873	−8.5***	−29.3*	20.0***	0.9198	−4.5	−67.6	−11.1	26.0	XGBoost	SRW
2017	7,994	0.0923	−5.7***	−13.1***	31.4**	0.9089	−4.2	−10.8	71.0	17.9	XGBoost	XGBoost
2018	8,512	0.1069	−4.6***	−6.5***	13.6***	0.3592	−1.5	−2.2	7.1	13.8	XGBoost	XGBoost
2019	9,023	0.1250	−9.9***	−11.9***	22.5***	0.4297	−5.1	−6.6	23.8	18.2	XGBoost	XGBoost
2020	9,606	0.1282	−8.7***	−10.1***	17.1***	0.4593	−4.7	3.1	10.2	13.3	XGBoost	XGBoost
2021 (Q1)	840	0.0899	−8.4***	−17.1***	14.9***	0.1618	−2.7	−25.4	15.1	15.4	XGBoost	XGBoost

5 Firm-level heterogeneity

Firm-level comparisons use the 2,124 firms with at least 12 common-sample forecasts (three years). Table 5 and Figures 3–4 show that XGBoost's gain is broad-based. It beats the SRW for 87.7% of firms by MAE (86.8% by RMSE), and the median firm-level improvement is 18.7%. Sign tests reject equal proportions for all three models. Drift and Foster beat the SRW for only 17.7% and 21.3% of firms. The aggregate result is not driven by a few firms. Excluding the 5% of firms that contribute most to XGBoost's MAE reduction still leaves a 18.2% improvement, and excluding the 5% hardest-to-forecast firms leaves 19.8%. The concentration curve in Figure 4(b) rises smoothly for XGBoost, while drift and Foster lose accuracy across most of the cross-section.

Table 5: Firm-level accuracy (firms with ≥12 common-sample forecasts). Firm %ΔMAE compares each model with the SRW on that firm's forecasts. The last two columns re-compute the pooled %ΔMAE after excluding the 5% of firms with the largest contribution to the model's MAE reduction, or the 5% of firms with the highest SRW MAE.

	Firm-level MAE			Firm %ΔMAE		% firms beating SRW		Pooled %ΔMAE excl.	
Model	p10	p50	p90	median	IQR	MAE	RMSE	top-5% contrib.	5% hardest
SRW	0.0205	0.0566	0.1664	–	–	–	–	–	–
SRW + drift	0.0209	0.0605	0.1836	−7.1	[−11.9, −2.3]	17.7	15.9	−9.8	−8.0
Foster	0.0208	0.0618	0.1908	−10.6	[−17.9, −2.0]	21.3	21.6	−17.6	−12.0
XGBoost	0.0177	0.0472	0.1299	18.7	[8.6, 27.2]	87.7	86.8	18.2	19.8

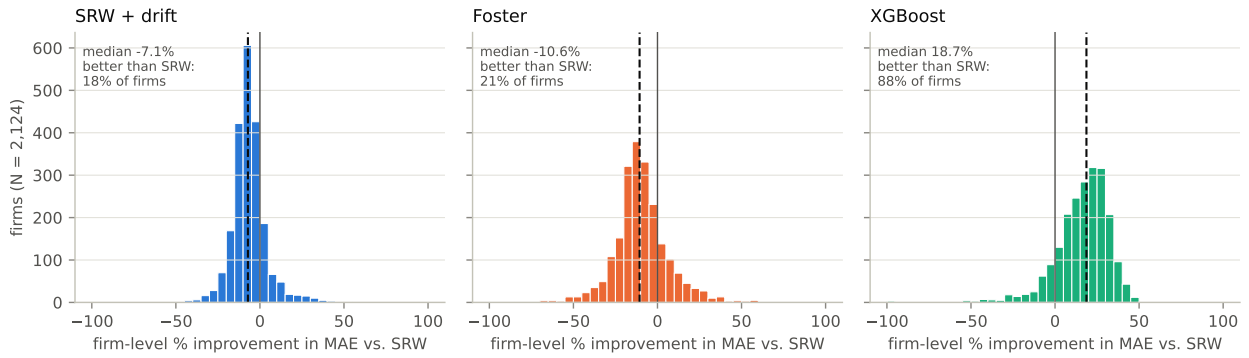


Figure 3: Distribution of firm-level percentage MAE improvement over the SRW (values clipped to ±100%; dashed line: median).

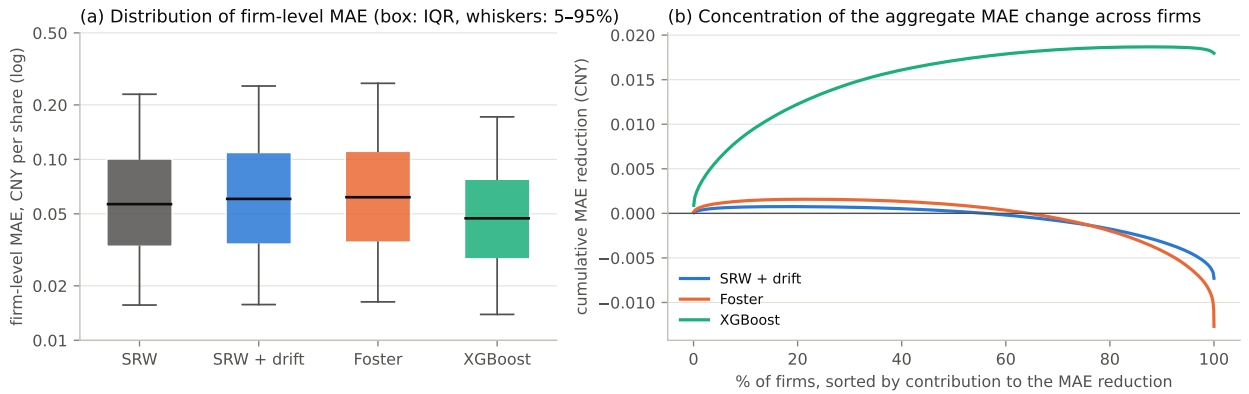


Figure 4: (a) Firm-level MAE by model. (b) Cumulative contribution of firms, sorted by their contribution, to the aggregate MAE reduction relative to the SRW (end point = aggregate MAE change).

6 Discussion and conclusion

Among the four models, only the pooled XGBoost model improves on the seasonal random walk. The gain is economically meaningful: about a fifth of the SRW's absolute forecast error, with a lower average absolute error in 95.4% of fiscal quarters. It is stable over 2005–2021 and spread across the large majority of firms. It does not come from model complexity per se. The two classical refinements, drift and Foster's autoregression, *reduce* accuracy, because 12-quarter firm-level histories cannot estimate weakly persistent dynamics precisely. XGBoost instead pools information across all firms. It also learns non-linear patterns that firm-level linear models cannot represent. The most important feature is a loss at $q-4$ (19.1% of total gain), followed by the seasonal lag EPS_{q-4}, the fiscal quarter, and the most recent YoY change. This points to strong reversal of losses, plausibly linked to Chinese delisting and special-treatment rules for loss firms, and to quarter-specific mean reversion that the SRW ignores.

For constructing alternative SUE measures, three implications follow.

1. $UE^{(\mathrm{XGB})}$ is a statistically better proxy for the unpredictable part of earnings than the SRW surprise. A drift- or Foster-based SUE is not recommended for this sample.
2. Because the forecast is more accurate, $UE^{(\mathrm{XGB})}$ is smaller in magnitude, and its scaling ($\sigma_i(UE)$) must be re-estimated from the model's own historical errors rather than taken from the SRW.
3. A more accurate forecast is not necessarily the market's expectation (Campbell et al., 2026). Whether PEAD and streak effects strengthen or weaken under $UE^{(\mathrm{XGB})}$ is itself informative: a stronger drift would indicate that investors anchor on naive seasonal comparisons.

The analysis has limitations:

- EPS is adjusted retroactively for capital changes and may contain later restatements.
- Pre-2004 dates are backfilled, which conservatively shrinks early training data.
- Same-day Q4/Q1 releases shorten the effective information set for a rising share of Q1 targets.
- The XGBoost grid is intentionally small; a richer model could change its accuracy but would not change the leakage-free design.

References

Bernard, V. L. and J. K. Thomas (1990). Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13(4), 305–340. https://doi.org/10.1016/0165-4101(90)90008-R

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics* 29(2), 238–249.

Campbell et al. (2026). Expectations matter: When (not) to use machine learning earnings forecasts. *Management Science*. https://doi.org/10.1287/mnsc.2024.05808

Chen, T. and C. Guestrin (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13(3), 253–263.

Foster, G. (1977). Quarterly accounting data: Time-series properties and predictive-ability results. *The Accounting Review* 52(1), 1–21.

Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55(3), 703–708.

Pagach, D. and R. Warr (2020). Analysts versus time-series forecasts of quarterly earnings: A maintained hypothesis revisited. *Advances in Accounting* 51, 100497. https://doi.org/10.1016/j.adiac.2020.100497

Thompson, S. B. (2011). Simple formulas for standard errors that cluster by both firm and time. *Journal of Financial Economics* 99(1), 1–10.

Zhang, Zhu, and Linnainmaa (2025). Man versus machine learning revisited. *Review of Financial Studies*. https://doi.org/10.1093/rfs/hhaf066